logweb Calculation Guide

How calories, XP, levels, and streaks are calculated in the fitness tracking system

Purpose: This guide explains the calculation logic in simple steps, so you can show it to a friend or classmate.

Important: Calorie values are estimated. logweb uses formulas and activity metadata, not medical or lab equipment.

1. Big Picture

When a user logs a workout, logweb does not ask the user to manually guess calories or XP. The backend calculates them automatically using real workout data.

• **Calories:** estimated from activity MET value, duration, user weight, distance, and exercise metadata.

• **XP:** rewards completion, time, intensity, performance, and streak consistency.

• **Level:** calculated from total accumulated XP.

• **Streak:** tracks consistency across weeks. A week is successful when the user completes at least 3 workouts.

Feature	Calculated from	Why it matters
Calories	MET, weight, minutes, distance, and volume	Shows estimated energy burned
XP	Completion, duration, MET, performance, and streak	Gamifies real effort without asking for manual XP
Level	Total XP	Shows long-term progress
Streak	Weekly workout consistency	Encourages habit building

2. Calorie Calculation

Calories are estimated mainly using MET. MET means Metabolic Equivalent of Task. A higher MET means the activity is more intense.

```
Basic formula:
calories = MET * 3.5 * weightKg / 200 * minutes
```

The Activity Library stores metadata for each activity, including default MET and calorie method. This lets Running, Walking, Cycling, Swimming, Strength Training, Yoga, and Sports use more realistic values.

2.1 Inputs Used for Calories

Input	Example	Used for
User weight	70 kg	Calories depend on body weight
Duration	40 minutes	Most formulas need time
Activity MET	Running = 9.8	Higher MET = higher burn rate
Distance	10 km	Cardio distance-only or pace-adjusted estimation
Strength sets/reps/weight	10 sets × 10 reps × 50 kg	Volume-based adjustment
Bodyweight factor	Push Up = 0.65	Better bodyweight strength estimation

2.2 Cardio Calorie Rules

• If time and distance are available, the backend can adjust MET based on pace.

• If only distance is available, it estimates from distance, body weight, and distance multiplier.

• If only time is available, it falls back to MET × duration × body weight.

```
Distance-only cardio example:
calories = distanceKm * weightKg * distanceMultiplier
```

2.3 Strength Calorie Rules

Strength training starts from MET, duration, and body weight, then adds a small modifier based on total training volume.

```
base = MET * 3.5 * weightKg / 200 * minutes
volume = sets * reps * weight
finalCalories = base * volumeModifier
```

For bodyweight movements, the app can use bodyweightFactor. Example: Push Up = 0.65, Pull Up = 1.0, Plank = 0.3.

2.4 Calorie Example: Running

Workout: Running, 40 minutes, 70 kg body weight, MET 9.8.

```
calories = 9.8 * 3.5 * 70 / 200 * 40
calories = 480.2
result ≈ 480 kcal
```

2.5 Calorie Example: Strength

Workout: 45 minutes strength training, 70 kg body weight, MET 6.0.

```
base = 6.0 * 3.5 * 70 / 200 * 45
base = 330.75 kcal
If volume is high, modifier may be 1.05 to 1.15
Example final = 330.75 * 1.05 ≈ 347 kcal
```

3. XP Calculation

XP is not the same as calories. Calories depend partly on body weight, so using calories as XP would be unfair. Instead, logweb XP rewards effort and consistency.

```
finalXp = round(baseCompletionXp + durationXp + intensityXp + performanceBonus +
streakBonus)
```

3.1 XP Components

Component	Formula	Maximum / Notes
Base completion XP	20	Given for completing a valid workout
Duration XP	minutes × 1.2	Capped at 90
Intensity XP	defaultMet × minutes × 0.15	Capped at 60
Cardio bonus	distanceKm × 4	Capped at 40
Strength bonus	totalVolumeKg / 500	Capped at 40
Bodyweight bonus	reps × bodyweightFactor × 0.25	Capped at 40
Streak bonus	Based on streak count	0, 10, 15, or 25

The backend uses the largest applicable performance bonus: cardio, strength, or bodyweight. This avoids over-rewarding one workout by stacking all performance types together.

```
performanceBonus = max(cardioBonus, strengthBonus, bodyweightBonus, 0)
```

3.2 Streak Bonus for XP

Streak count	Bonus XP
0–1	0 XP
2–3	10 XP
4–6	15 XP
7+	25 XP

3.3 XP Example: Running

Workout: Running, 10 km, 40 minutes, MET 9.8, streak = 0.

```
baseCompletionXp = 20
durationXp = min(40 * 1.2, 90) = 48
intensityXp = min(9.8 * 40 * 0.15, 60) = 58.8
performanceBonus = min(10 * 4, 40) = 40
streakBonus = 0
finalXp = round(20 + 48 + 58.8 + 40 + 0) = 167 XP
```

3.4 XP Example: Strength

Workout: 45 minutes strength training, MET 6.0, 10 sets × 10 reps × 50kg = 5000kg total volume, streak = 3.

```
baseCompletionXp = 20
durationXp = min(45 * 1.2, 90) = 54
intensityXp = min(6.0 * 45 * 0.15, 60) = 40.5
performanceBonus = min(5000 / 500, 40) = 10
streakBonus = 10
finalXp = round(20 + 54 + 40.5 + 10 + 10) = 135 XP
```

4. Level Calculation

Level is calculated from total XP. Early levels are easier, while later levels require more XP.

```
xpNeededForNextLevel = 100 + (level * 75) + (level^2 * 15)
```

The backend stores cumulative level thresholds. Example: Level 1 starts at 0 XP, Level 2 starts at 190 XP, Level 3 starts at 500 XP, Level 4 starts at 960 XP.

```
currentLevelXp = totalXp - cumulativeXpRequiredForCurrentLevel
```

Example: If Level 3 starts at 500 XP and a user has 620 total XP, then currentLevelXp = 620 - 500 = 120 XP progress inside Level 3.

5. Streak Calculation

The backend streak system is weekly. It rewards consistency, not only one intense day. A completed week counts when the user logs at least 3 workouts during that week.

5.1 Weekly Streak Steps

• The app uses Cambodia local week boundaries: Monday to Sunday.

• At the start of a new week, the backend checks the previous completed week.

• If the previous week has 3 or more workouts, weekly streak increases by 1.

• If the previous week has fewer than 3 workouts, the streak becomes at risk or broken depending on restore rules.

• The app tracks longest weekly streak separately from current weekly streak.

5.2 Streak Restore Logic

• If the streak is at risk, the user may restore it with a free streak freeze if available.

• If no freeze is available, the user may restore using XP if allowed and if they have enough XP.

• Restore cost is calculated as: min(150, 50 + 10 × weeklyStreak).

• If the restore deadline is missed, the streak resets to 0.

```
restoreCost = min(150, 50 + 10 * weeklyStreak)
```

5.3 Weekly Streak Example

Week	Workout count	Result
Week 1	3 workouts	Streak becomes 1
Week 2	4 workouts	Streak becomes 2
Week 3	2 workouts	Streak becomes at risk
Week 3 restored	Freeze or XP restore used	Streak stays alive
Missed restore deadline	No restore	Streak resets to 0

6. Why This System Is Fair

• Calories are estimated from real activity science using MET and user data.

• XP is based on effort and consistency, not just body weight.

• Strength, cardio, bodyweight, yoga, and sports can all earn XP fairly.

• The system saves xpBreakdown, so the app can explain why the user earned a certain XP amount.

• Streak focuses on habit building: at least 3 workouts per week.

7. Short Presentation Script

You can say this when showing the system:

```
In logweb, users do not type calories or XP manually.
When a workout is saved, the backend calculates calories from MET, user weight, duration,
distance, and activity metadata.
Then it calculates XP from completion, duration, intensity, performance, and streak
bonus.
The app also tracks levels from total XP and weekly streaks from workout consistency.
This makes the fitness tracking more realistic and harder to fake from the frontend.
```

End of guide